

Vertex Operators and Symmetric Functions

Introduction to Modern Advances in Algebra

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Heisenberg algebra: $\mathfrak{h} = \text{span}\{a_n\}_{n \in \mathbb{Z}}$

$$[a_n, a_m] = n\delta_{n+m,0}$$

a_n are the modes of the affine vertex algebra associated to $\mathfrak{gl}_1$

Part of the data of this vertex algebra is its vertex operator:

$$V(z) = \exp\left(\sum_{n>0} \frac{1}{n} z^n a_{-n}\right) \exp\left(\sum_{n>0} \frac{1}{n} z^{-n} a_n\right)$$

Symmetric Functions

Λ : Ring of symmetric functions in $x_1, x_2, \dots$

$$p_n = x_1^n + x_2^n + \dots \text{ (power sums)}$$

$$\Lambda \cong \mathbb{C}[p_1, p_2, \dots]$$

$$\mathfrak{h} \simeq \Lambda: \text{ for } n \in \mathbb{Z}_{>0}$$

$$a_{-n}f = p_n f, \quad \text{and} \quad a_n f = n \frac{\partial}{\partial p_n} f.$$

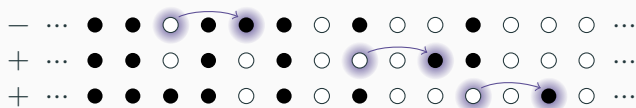
Fermionic Fock Space

Maya diagram: row of black and white dots (“go stones”) with appropriate boundary conditions, eg



$\mathcal{F}$ is the span of all such diagrams

$\mathfrak{h} \curvearrowright \mathcal{F}$, eg, acting on diagram above with a_{-2} gives



Theorem (Boson-Fermion Correspondence)

The $\mathfrak{h}$ -modules Λ and $\mathcal{F}$ are isomorphic

Vertex Operators

$$h_n = \sum_{i_1 \leq \dots \leq i_n} x_{i_1} \cdots x_{i_n} \text{ (complete homogeneous)}$$

$$h_1 = x_1 + x_2 + \dots, h_2 = x_1^2 + x_2^2 + x_1 x_2 + \dots$$

$$\text{Generating function: } H(z) = \sum_{n \geq 0} z^n h_n = \prod_{i \geq 1} \frac{1}{1 - zx_i}$$

By identities between symmetric functions

$$H(z) = \exp \left(\sum_{n > 0} \frac{1}{n} z^n p_n \right)$$

which corresponds to the action of the $\mathfrak{h}$ half vertex operator

$$\exp \left(\sum_{n > 0} \frac{1}{n} z^n a_{-n} \right)$$

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